

Mathematics Made Easy With Minimal Octave

This fifth article in the mathematical journey through open source introduces Octave, a non-programmer's way of doing mathematics.



Octave sounds like a musical note, but in reality it is the name of its author's Chemical Engineering professor, who was known for his 'back-of-the-envelope calculations'.

All of bench calculator, and more!

All the operations of the bench calculator (*bc*) are just a subset of *octave*. So, there's no point in reinventing the wheel. Whatever operations can be done in *bc*—arithmetic, logical, relational and conditional—can also be done with equal ease in *octave*. And like *bc*, it can be used as a mathematical programming language. So, why did we waste time learning *bc*? We might as well have come directly to *octave*. Well, there's one difference—precision. You can't get that in *octave*. If you need it, you would have to go back to *bc*. With *octave*, let's start with N-dimensions, or in other words, vectors and matrices.

Getting started

Commands: Typing '*octave*' on the shell brings up the *octave*'s interactive shell. Type 'quit' or 'Control-D' to exit. The interactive shell starts with a welcome message. As in *bc*, use the option *-q* for it not to show. Additionally, use the *-f* option for it not to pick up any local start-up scripts (if any). So, for our examples, the command would be '*octave -qf*' to start *octave* with an interactive shell.

Prompts and results: The input prompt is typically denoted by '*octave:X>*', where X is just a number showing the command count. Valid results are typically shown with '*<variable> =*', or '*ans =*' or '*=>*'. The command count is incremented for the next input. Errors cause error messages to be printed and then *octave* brings back the input prompt without incrementing the command count. Putting a semicolon (;) at the end of a statement

suppresses the result (return value) to be displayed.

Comments could start with # or %. For block comments, #[...]# or %{ ... }% can be used. Detailed help on a particular topic can be obtained using 'help <topic>' and the complete documentation can be accessed using 'doc' on the *octave* shell.

```
$ octave -qf
octave:1> # This is a comment
octave:1> pi # Built-in constant
ans = 3.1416
octave:2> e # Built-in constant again
ans = 2.7183
octave:3> i # Same as j - the imaginary number
ans = 0 + 1i
octave:4> x = 3^4 + 2^30;
octave:5> x
x = 141
octave:6> y
error: 'y' undefined near line 6 column 1
octave:6> doc # Complete doc; Press 'q' to come back
octave:7> help plot # Help on plot
octave:8> A = [1 2 3; 4 5 6; 7 8 9] # 3x3 matrix
A =

    1    2    3
    4    5    6
    7    8    9

octave:9> quit
```

Matrices with a heart

We have already seen matrix creation. They can also be created through multiple lines. *Octave* continues waiting for